THE INTERNAL HOM COMPLEX IS WELL DEFINED UP TO QUASI-ISOMORPHISM IN BOTH VARIABLES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: both H^0 groups are computed and shown equal, which is exactly the asserted quasi-isomorphism after the usual reading of the displayed statement.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology for a ringed space. That statement is the special case of this ringed-site statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The single labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer to the same chapter.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- SheafHom complex of complexes of O-modules and its cohomology sheaves. Same chapter, the construction of SheafHom^bullet for complexes of modules on a ringed site. Supplies the two complexes compared in the statement and the H^0 groups computed in the proof.
- K-injective complex of O-modules. Same chapter, the definition of K-injectivity and its use in computing derived homs. Supplies the hypothesis that both I^bullet and (I')^bullet are K-injective, which is what makes the H^0 groups computable.
- Sheaf associated to the presheaf U -> Hom_{D(O_U)}(L|_U, M|_U). Same chapter, the description of the sheaf of homomorphisms in the derived category by restriction to O_U. Supplies the common value of the two H^0 groups, from which quasi-isomorphism follows.

Source reference. Original lemma 0A94: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 8720–8720. Same-chapter navigation pointer invoked at proof line 8774, where the author identifies both H^0 groups with the same sheafified presheaf of derived homs. The pointer only resolves that reference for the reader; the cited result is not reproduced inside the counted span, and no bibliography citation is involved.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $(\mathcal{I}')^\bullet \to \mathcal{I}^\bullet$ be a quasi-isomorphism of K-injective complexes of $\mathcal{O}$-modules. Let $(\mathcal{L}')^\bullet \to \mathcal{L}^\bullet$ be a quasi-isomorphism of*

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complexes of $\mathcal{O}$-modules. Then

$$\mathcal{H}om^\bullet(\mathcal{L}^\bullet, (\mathcal{I}')^\bullet) \longrightarrow \mathcal{H}om^\bullet((\mathcal{L}')^\bullet, \mathcal{I}^\bullet)$$

is a quasi-isomorphism.

Proof. Let M be the object of $D(\mathcal{O})$ represented by $\mathcal{I}^\bullet$ and $(\mathcal{I}')^\bullet$. Let L be the object of $D(\mathcal{O})$ represented by $\mathcal{L}^\bullet$ and $(\mathcal{L}')^\bullet$. By Lemma 0A94 we see that the sheaves

$$H^0(\mathcal{H}om^\bullet(\mathcal{L}^\bullet, (\mathcal{I}')^\bullet)) \quad \text{and} \quad H^0(\mathcal{H}om^\bullet((\mathcal{L}')^\bullet, \mathcal{I}^\bullet))$$

are both equal to the sheaf associated to the presheaf

$$U \longmapsto \mathrm{Hom}_{D(\mathcal{O}_U)}(L|_U, M|_U)$$

Thus the map is a quasi-isomorphism. □